

Sri Lanka Institute of Advanced Technological Education

Advanced Technological Institute – Gampaha

Software Requirements Specification (SRS)



**Development of a Web-Based Learning Management System (LMS) For
Aura Pen Saku Education Institute**

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GAM/IT/2324/F/0099

Date of Submission: **08th May 2026**

Table of Content

1.	Introduction	1
1.1	Purpose	1
1.2	Scope	1
1.3	Definitions, Acronyms, Abbreviations	1
1.4	References	1
2.	Overall Description	2
2.1	Product Perspective.....	2
2.2	Product Functions.....	2
2.3	User Classes and Characteristics	2
2.4	Operating Environment.....	2
2.5	Design and Implementation Constraints	2
2.6	Assumptions and Dependencies.....	2
3.	External Interface Requirements	3
3.1	User Interfaces.....	3
3.2	Hardware Interfaces	3
3.3	Software Interfaces	3
3.4	Communication Interfaces.....	4
4.	System Features (Functional Requirements)	5
4.1	Feature 1: Role-Based User Authentication	5
4.2	Feature 2: Grade-Based Content Management	5
4.3	Feature 3: Interactive Activity Module.....	5
4.4	Feature 4: Student Progress Tracking & Motivation	5

5. Non-Functional Requirements	6
5.1 Performance Requirements	6
5.2 Security Requirements	6
5.3 Usability Requirements.....	6
5.4 Reliability Requirements	6
5.5 Scalability Requirements	7
6. Other Requirements.....	7
6.1 Database Requirements	7
6.2 Legal Requirements.....	7
7. Appendices.....	8
7.1 Sample Screens (Optional)	8
7.2 Future Enhancements.....	8
8. Approval.....	8

1. Introduction

1.1 Purpose

The purpose of this document is to specify the comprehensive software requirements for the "Development of a Web-Based Learning Management System (LMS) for Aura Pen Saku Education Institute." This document is intended for project supervisors, developers, and stakeholders to ensure a common understanding of the system's functionality.

1.2 Scope

This system is designed to digitalize the academic operations of Aura Pen Saku Institute. The primary focus is providing an organized platform for Grade 9 students to access ICT (Information and Communication Technology) materials. The system will bridge the gap between teachers and students by providing online video lessons, downloadable tutes, interactive activities ,PDF notes and automated MCQ assessments.

1.3 Definitions, Acronyms, Abbreviations

- SRS: Software Requirements Specification
- LMS: Learning Management System
- UI: User Interface
- MCQ: Multiple Choice Questions
- API: Application Programming Interface

1.4 References

- Aura Pen Saku Education Institute's curriculum.
- IEEE Standard for Software Requirements Specification
- Gemini AI
- React official Documentation
- Node.js API References
- MYSQL 8.0 Reference manual

2. Overall Description

2.1 Product Perspective

This is a new standalone web application. It follows a three-tier architecture using React.js (Frontend), Node.js (Backend), and MySQL (Database).

2.2 Product Functions

- User registration and role-based secure login.
- Displaying ICT curriculum organized by chapters.
- Video streaming and PDF document downloading.
- Interactive learning activities (Matching, Fill-in-the-blanks).
- Student progress tracking and performance reporting
- subject-wise lesson management

2.3 User Classes and Characteristics

- Admin: Full control over the system, including managing students, teachers, and adding new subjects.
- Teacher: Responsible for uploading notes and creating interactive activities for their subject.
- Student: Can view lessons, download notes, and perform quizzes and activities.

2.4 Operating Environment

- The system will run on any modern web browser (Google Chrome, Firefox, Edge).
- It requires Node.js runtime environment on the server.

2.5 Design and Implementation Constraints

- The system must be developed using React.js and Node.js.
- It must follow a responsive design to work on different screen sizes (Laptop/Desktop).
- Limited to educational content (non-commercial)

2.6 Assumptions and Dependencies

It is assumed that the institute will provide the necessary lesson content and that users have access to the internet.

3. External Interface Requirements

3.1 User Interfaces

- Landing Page: A professional web interface introducing Aura Pen Saku Institute with dedicated login portals for Administrators, Teachers, and Students.
- Student Dashboard: A React-based interactive dashboard displaying the student's current grade, enrolled subjects (e.g., Grade 9 ICT), and a graphical progress bar for lesson completion.
- Content Management Interface (CMS): A specialized interface for Teachers to upload, categorize, and update YouTube video links and PDF course materials based on grade and subject.
- Interactive Learning Module: A high-engagement UI designed for end-of-lesson activities such as "Matching Games" and "Fill-in-the-blanks" with real-time feedback animations.

3.2 Hardware Interfaces

- Client Devices: The system is optimized for Desktops and Laptops with a minimum requirement of 4GB RAM to ensure smooth rendering of the React-based frontend.
- Server Side: In accordance with the proposal, a system with a 13th Gen Intel Core i5 processor and 16GB RAM will be utilized for backend operations and database hosting.

3.3 Software Interfaces

- Frontend Technology: React.js (v18+) used for building a responsive and dynamic User Interface.
- Backend Framework: Node.js with Express.js for handling server-side routing, business logic, and API management.
- Database Management: MySQL 8.0 utilized for structured relational data storage including student records and academic content.
- Operating Systems: Compatible with Windows 11 Pro, Linux, and macOS environments via modern web browsers (Chrome/Edge).

3.4 Communication Interfaces

- HTTP/HTTPS Protocols: Used for secure data transmission between the React client and Node.js server.
- RESTful APIs: The system architecture relies on REST principles for efficient CRUD (Create, Read, Update, Delete) operations.
- JSON Data Exchange: All data communicated between the frontend and backend is formatted in JSON (JavaScript Object Notation).

4. System Features (Functional Requirements)

4.1 Feature 1: Role-Based User Authentication

- Description: This feature allows Students and Administrators to log into the system securely. It identifies the user role and provides access to the relevant dashboard while protecting user data.
- Inputs: Registered Username or Email and Password.
- Outputs: Successful redirection to the Student Dashboard or Administrative Control Panel.

4.2 Feature 2: Grade-Based Content Management

- Description: This feature enables Teachers to manage educational resources. They can upload YouTube lesson links and PDF handouts, categorizing them by grade and subject.
- Inputs: YouTube Video URL, PDF file upload, Subject title, and Grade category.
- Outputs: Updated lesson repository visible to students on their personalized dashboard.

4.3 Feature 3: Interactive Activity Module

- Description: An engaging module where students can participate in ICT-related activities such as Matching games, Fill-in-the-blanks, and MCQ quizzes to test their knowledge.
- Inputs: Student's selected answers or text inputs for the activities.
- Outputs: Automated marks calculation and immediate performance feedback for the student.

4.4 Feature 4: Student Progress Tracking & Motivation

- Description: This feature monitors the student's learning journey. It tracks completed lessons and awards "Star Badges" as rewards for finishing tasks, which motivates the student to continue learning.
- Inputs: Records of completed video lessons and passed assessment scores.
- Outputs: Visual progress bar updates and a display of earned badges on the student profile.

5. Non-Functional Requirements

5.1 Performance Requirements

- **Response Time:** The system shall be highly responsive, with all page loads and database queries (e.g., retrieving ICT lessons) completing in less than 3 seconds.
- **Throughput:** The backend (Node.js) shall handle multiple concurrent student requests efficiently without performance degradation.

5.2 Security Requirements

- **User Authentication:** Access to the LMS is strictly limited to registered users. Password-protected login is mandatory for Students, Teachers, and Admins.
- **Data Encryption:** All sensitive information, especially user passwords, shall be hashed using the BCrypt algorithm before being stored in the MySQL database.
- **Session Management:** The system shall implement secure session handling to prevent unauthorized access to user accounts.

5.3 Usability Requirements

- **User-Friendly Interface:** The UI shall be designed using Bootstrap 5 to ensure a clean, intuitive, and attractive layout suitable for Grade 10 and 11 students.
- **Responsiveness:** The system shall be fully responsive, providing a consistent experience across various screen sizes (Desktops, Laptops, and Tablets).
- **Navigation:** The navigation flow shall be simple, allowing students to access any lesson or activity within a maximum of three clicks.

5.4 Reliability Requirements

- **System Uptime:** The system shall aim for a 99% uptime to ensure students can access educational materials whenever needed.
- **Data Consistency:** The MySQL database shall maintain strict referential integrity to prevent data loss or corruption during updates.

5.5 Scalability Requirements

- **User Capacity:** As specified in the proposal, the system architecture shall be capable of supporting up to 500 concurrent users without affecting stability.
- **Content Growth:** The system shall allow for an easy increase in the volume of video and PDF content as the institute grows.

6. Other Requirements

6.1 Database Requirements

- **Database Type:** The system will utilize MySQL 8.0, which is a Relational Database Management System (RDBMS).
- **Data Storage:** Centralized storage will be maintained on the server to store:
- **User Records:** Student and Teacher credentials, profile information, and role-based permissions.
- **Course Content:** Metadata for YouTube video links, PDF file paths, and subject/grade categorizations.
- **Assessment Data:** Question banks for MCQ quizzes and interactive activity scores.
- **Progress Logs:** Detailed records of student lesson completion and earned "Star Badges."
- **Data Integrity:** Primary keys and Foreign keys will be used to ensure relationships between tables (e.g., linking StudentID to Activity Results) remain consistent.

6.2 Legal Requirements

- **Data Protection:** The system will ensure that all student and teacher data is kept private and secure, following standard data protection practices.
- **Content Copyright:** Aura Pen Saku Institute is responsible for ensuring that all uploaded PDF handouts and video materials comply with educational copyright regulations.
- **Academic Integrity:** The assessment module will be designed to prevent unauthorized access to answer keys before a student completes an activity.

7. Appendices

7.1 Sample Screens (Optional)

The following represent the planned user interface designs for the Aura Pen Saku LMS. These mockups illustrate the layout and navigation flow for the primary users (Students and Teachers).

- Student Login Page: A clean interface for students to securely enter their credentials.
- Student Dashboard: Featuring the Grade 9 ICT lesson categories, progress tracking bars, and the "Star Badges" reward section.
- Lesson Viewer: A split-screen interface displaying the YouTube video player and PDF handout side-by-side.
- Interactive Activity Module: A sample mockup of the "Matching Game" and "Fill-in-the-blanks" interface.

7.2 Future Enhancements

- Mobile Application: Developing a dedicated mobile app for Android and iOS platforms.
- Live Chat Support: Implementing a real-time chat module for students to communicate with teachers directly.
- E-Certification: Automated generation of completion certificates after finishing a grade curriculum.

8. Approval

Name	Role	Signature	Date
N.D. Gunasekara	Supervisor		
P.P.K.M. Gunathilaka	Developer		

